University of Southern Denmark

EXAM

Written Exam

AI504: Knowledge Representation

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Allowed aids:	No aids
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DEPARTMENT OF MATHEMATICS
AND COMPUTER SCIENCE

Contents

Multiple choice questions

1

(a) ×
(b) ✓
(c) ×
(d) ×

2

(a) ✓
(b) ×
(c) ×
(d) ×

3

(a) ×
(b) ×
(c) ×
(d) ✓

4

(a) ×
(b) ✓
(c) ×
(d) ×

5

(a) ✓
(b) ×
(c) ×
(d) ×

6

(a) ×
(b) ✓
(c) ×
(d) ×

7

(a) ×
(b) ×

(c) ×
(d) ✓

8

(a) ×
(b) ×
(c) ✓
(d) ×

9

(a) ×
(b) ✓
(c) ×
(d) ×

10

(a) ×
(b) ×
(c) ×
(d) ✓

11

(a) ✓
(b) ×
(c) ×
(d) ×

12

(a) ×
(b) ×
(c) ✓
(d) ×

Open questions

1

Given that

$$(e \iff \neg b) \wedge (b \iff \neg d) \wedge (d \iff \neg a) \wedge (a \iff \neg c) \wedge (c \iff \neg e)$$

This means that $e \iff \neg e$ which mean that there is **0** solutions

2

Lets prove that

$$(\forall x \in [\![c]\!])(\forall y \in [\![r \text{ all } d]\!])(x, y) \in [\![r]\!]$$

Because of the transitive relation where $\forall x, y, z \in M$ if $(x, y) \in [\![r]\!] \wedge (y, z) \in [\![r]\!]$ then $(x, z) \in [\![r]\!]$

This mean that $[\![d]\!] \subseteq [\![r \text{ all } d]\!]$. Therfore

$$(\forall x \in [\![c]\!])(\forall y \in [\![r \text{ all } d]\!])(x, y) \in [\![r]\!] \Rightarrow (\forall x \in [\![c]\!])(\forall y \in [\![d]\!])(x, y) \in [\![r]\!]$$

i.e

$$\text{All } c \ (r \text{ all } (r \text{ all } d)) \Rightarrow \text{All } c \ (r \text{ all } d)$$

$\square$

Now lets construct a model $\mathcal{N}$ which does *not* satisfy this.

Let $\mathcal{N}$ have the domain $M = \{1, 2\}$ Then

$$[\![c]\!] = \{1\}, \quad [\![c]\!] = \{2\}, \quad \text{and} \quad [\![r]\!] = \{(1, 2)\}$$

This means that

$$[\![r \text{ all } d]\!] = \{1\}$$

$$[\![r \text{ all } (r \text{ all } d)]\!] = \emptyset$$

This proves that

$$\mathcal{N} \models \text{All } c \ (r \text{ all } (r \text{ all } d))$$

And also that

$$\mathcal{N} \not\models \text{All } c \ (r \text{ all } d))$$

$\square$

3

The property is:

Any sentence ϕ which is not an axiom, must be either be a single letter, or a sentence on the form: All $(r \text{ all } \psi)\ \rho$ where ψ and ρ is arbitrary sentences with the same property.

- All $(r \text{ all } c)\ d$ must have this term by definition
- All $c\ (r \text{ all } d)$ must *not* have this term by definition
- Since no axioms are allowed under this property, no sentences on the form All $x\ x$ has this property.
- Lets try and prove this last one using induction

Proof by induction

Goal: if $\Gamma \vdash \phi$ and ϕ has the property, then there must exists a sentence in Γ which has the property

1 **<u>Base case</u>**

2 Let $\Gamma = \{\phi\}$,

3 Then since ϕ has this property, and since $\phi \in \Gamma$, there is an element in ϕ which has this property.

4 **<u>Inductive hypothesis</u>**

5 Since $\Gamma \vdash \phi$

6 Assume that by applying the logic rules Barbara and down, one can derive ϕ from Γ

7 **<u>Inductive step</u>**

8 By ***IH***, one can apply both Barbara and down to derive ϕ

Barbara

9
$$\frac{\text{All } x\ y \qquad \text{All } y\ z}{\text{All } x\ z}$$

The Barbara does preserve this propery as it does not which around any terms.

10 Down

11
$$\frac{\text{All } x\ (r \text{ all y}) \qquad \text{All } z\ y}{\text{All } x\ (r \text{ all } z)}$$

12 Since Down preserves the property of ϕ, that is, it also does not switch around any terms (note that in the image x and y of course will not be a single letter as this would interfere with the property)

13 Because of this, deriving ϕ from Γ using these rules will find an element in Γ which has this rule.

4

For logic $\mathcal{A}$ since one can only use the Barbara rule, ϕ can only be axioms since the two sets of sentences have no overlap

$$\text{All } 1 \;\; 1, \text{All } 2 \;\; 2, \text{All } 3 \;\; 3$$

5

Let $\Gamma \vDash^f \phi$ be true if $(\forall p \in P) : [\![p]\!] \neq \emptyset$

Now let $\mathcal{N}$ be an arbitrary model.

If $\mathcal{N} \vDash \phi$ and $\mathcal{N} \nvDash^f \phi$. This means that $\exists p \in P : [\![p]\!] = \emptyset$ Lets now add som elements to the domain such that.

$$(\forall p \in P) : [\![p]\!] \neq \emptyset.$$

Since no element have been removed, $\mathcal{N} \vDash \phi$ must still hold as must $\mathcal{N} \nvDash^f \phi$

$\square$

6

(a) The canonical model $\mathcal{M}_\Gamma$ has the property $\mathcal{M}_\Gamma \vDash \phi \iff \Gamma \vdash \phi$

Since this logic is both sound and complete, the following is true

$$\Gamma \vDash \phi \Rightarrow \Gamma \vdash \phi \text{ and } \Gamma \vdash \phi \Rightarrow \Gamma \vDash \phi$$

Because of this both of the following are true

$$\Gamma \vDash \phi \Rightarrow \mathcal{M}_\Gamma \vDash \phi$$
$$\mathcal{M}_\Gamma \vDash \phi \Rightarrow \Gamma \vDash \phi$$

7

Lets prove this by induction

Proof by induction

Goal: if ϕ is a tautology, then ϕ^R is also a tautology.

1 **Base case**

2 Case 1 ($\phi = p$)

3 Let ϕ be a single sentence letter p

4 ϕ cannot be a tautology so this holds.

5 Case 2 ($\phi = p \Rightarrow q$)

6 Let ϕ be a sentence on the form $p \Rightarrow q$ where p, and q are single sentenced letters

7 ϕ can only be a tautology if $p = q$.

8 Then $(p \Rightarrow q)^R = q \Rightarrow p$, this is still a tautology. So it holds

9 **Inductive hypothesis**

10 For any subterm t where t can either be a single sentence letter or a sentence on the form $p \Rightarrow q$. if t is a tautology, then t^R is also a tautology.

11 **Inductive step**

12 Let ϕ be an arbitrary sentence consisting of subterms t_1 and t_2

13 $$\phi = t_1 \Rightarrow t_2$$

14 If both t_1 and t_2 are tautologies, then both $t_1 \Rightarrow t_2$ and $t_2 \Rightarrow t_1$ are tautologies.

15 Because of this if ϕ is a tautology, ϕ^R must be as well.

8

Both a, b must be less than c, d

$$\min(a, b) < \max(c, d) \Longleftrightarrow (a < c \land a < d) \lor (b < c \land b < d)$$